Rakshita Vijay

The best systems are the ones people never have to think about.

Chennai, India | Chennai, India | GitHub | LinkedIn

About

I'm a third-year Computer Science Engineering student at BITS Pilani, Hyderabad, with a habit of picking projects apart down to the data structure underneath - whether that's a dependency graph deciding the order database objects get migrated in, or a function-call graph deciding whether an Android app is malicious.

Most recently I was a Software Engineering Intern in Microsoft's Azure Data Migration team, where I extended an AI-assisted schema-conversion pipeline to handle SQL Server as a source: a topological chunk scheduler, a hybrid regex-plus-LLM translation router, and a fault-tolerant compile-and-retry loop that got ~97% of chunks converting without manual intervention. Before that, at CultureVo, I built a multi-agent conversational AI toolkit spanning six personas and four languages.

Outside of internships, I prompt-engineer AI-driven tools end to end - a finance tracker with a multi-agent CrewAI insight pipeline, a project-tagging workspace on Next.js and Supabase, an async research-and-ideation pipeline that cut a team's turnaround from five days to one sitting.

When I'm not at a keyboard, I'm usually working a crochet or knitting pattern - a different kind of graph traversal - or buried in a novel, mystery and/or fantasy being my usual picks.

Education

BITS Pilani, Hyderabad Campus *2023 - 2027*
B.E. Computer Science Engineering · CGPA 8.64

St. John's Public School *Class XII · 2023*
CBSE · 94.2%

Chettinad Vidyashram *Class X · 2021*
CBSE · 95.8%

Technical Skills

Python, C++, C, Java, SQL, PyTorch, Next.js, React, Supabase, Vercel, CrewAI, Gemini API, Prompt Engineering, Git

Experience

Microsoft - Azure Data Migration Team *May 2026 - Jul 2026*
Software Engineering Intern · SQL Server → PostgreSQL Schema Conversion Engine

- Extended an AI-assisted Oracle→PostgreSQL schema-conversion pipeline to support SQL Server as a source, with dialect-aware gating across a 4-stage Extract→Chunk→Convert→Verify pipeline.
- Built a dependency-aware chunk scheduler on Kahn's Algorithm to topologically order database objects, correctly handling cycles and self-referencing edges while upholding ACID transactionality.
- Designed a hybrid conversion router splitting mechanical DDL to a regex fast-path and procedural/semantic logic to LLM translation, sharing a canonical identifier map across both paths.
- Built a fault-tolerant compile-and-retry loop with auto-retry from compile-error feedback - ~97% of chunks converged without manual intervention.
- Stress-tested against Northwind, AdventureWorks, WideWorldImporters, and a custom 7-schema database: ~98% of in-scope objects across 45+ chunks (600+ objects), cutting runtime from ~4 hours to 35–45 minutes.

CultureVo *May 2025 - Jul 2025*
Summer Intern · Conversational AI Tools

- Built PersonaBot, a configurable multi-agent orchestration toolkit with a state-managed persona system spanning 6 profiles and 10 customizable traits.
- Implemented persistent, multi-session conversation state across four languages (English, Sinhala, Tamil, Hindi), enabling bulk generation of 100+ Q&A pairs.

Projects

Android Malware Detection with GNNs (GraphSAGE, Androguard, PyTorch) *Jan 2026 - May 2026*
Cybersecurity · Graph ML

- Built an end-to-end malware classification pipeline extracting Function Call Graphs via Androguard and training GraphSAGE models to distinguish benign from malicious Android apps.

- Designed an API-aware, structure-preserving graph pruning technique that removes low-signal non-API methods via weighted edge rewiring.
- Integrated pretrained Skip-Gram node embeddings with checkpoint-based training for resumable HPC runs.
- Achieved a 94.39% F1-score and 92.41% accuracy on an imbalanced dataset of 150K benign and 50K malicious samples.

Finance Tracker & Analyser (Next.js, CrewAI, Vercel) *Jun 2025 - Jul 2025*
AI-Driven Financial Analytics

- Designed a multi-format analysis and report-generation service backed by a 2-agent pipeline for automated insight generation, since restored to a fuller multi-agent CrewAI pipeline with live per-stage timers.
- Implemented real-time budget tracking with configurable spending categories and an AI-driven recovery-strategy module.
- Built a rules-based anomaly-detection service that flags inconsistencies in uploaded statement data.

Automated Research & Ideation Pipeline (Python, asyncio, Multi-Agent) *Jun 2025*
Multi-Agent Systems · Async Automation

- Designed a 5-agent orchestration workflow automating topic selection, research, and outline generation with shared agent state.
- Implemented concurrent agents over Python asyncio to generate 7 niche titles per theme, each with sourced research summaries and outline guides.
- Cut research-and-ideation turnaround by ~80% (5 days → 1 session); adopted as the team's standard ideation tool.

Tag Atlas (Next.js, Supabase, Gemini) *Personal project*
Project write-up storage & tagging tool

- Sidebar-plus-detail-panel workspace for storing and tagging project write-ups, with tag-based similarity ranking and an archive flow.
- AI-assisted tag suggestions via Gemini, password auth, and bulk project-insertion tooling.

Leadership

ACM - BPHC Chapter *Nov 2024 - Jul 2026*
Content & Research Team Co-Lead, Junior Office & Chair, Senior Office

- Authored 5+ technical articles for the ACM BPHC Blog and The BITSian Gazette (Aug–Nov '24 edition).
- Coordinated logistics as Point of Contact for 4 tech events, including a coding workshop for 70+ indigent students.

ELAS, BPHC *Dec 2024 - Jan 2025*
ARG Team Lead, Verba Maximus '25

- Led a 13-member team to design and execute a 6-hour event that drew 125+ participants - the highest-footfall headliner event at VM '25.

Talks & Workshops

Introduction to Machine Learning *Aug 2025*
Organized by ACM - BPHC Chapter

- Co-hosted a 250+ attendee workshop covering supervised/unsupervised learning, regression, neural networks, and activation functions with live coding.